

# Derivation of Wall Vorticity Boundary Conditions for the Streamfunction–Vorticity Formulation

## 1 Introduction

For the two-dimensional incompressible Navier–Stokes equations in streamfunction–vorticity form,

$$\nabla^2\psi = \omega \quad (1)$$

the streamfunction is known on the walls, but the wall vorticity is not known directly. To solve the vorticity transport equation and the Poisson equation for the streamfunction, we must determine the vorticity values on the boundaries. This is accomplished using a Taylor series expansion near each wall. The streamfunction definitions are

$$u = \frac{\partial\psi}{\partial y} \quad (2)$$

$$v = -\frac{\partial\psi}{\partial x} \quad (3)$$

and the Poisson equation is

$$\nabla^2\psi = \omega. \quad (4)$$

Throughout the cavity problem, the streamfunction is constant on all walls:

$$\psi_{wall} = 0. \quad (5)$$

## 2 Bottom Wall

The bottom wall is located at

$$j = 0$$

and the first interior node is

$$j = 1.$$

The wall is stationary, therefore

$$u = 0. \quad (6)$$

Using Taylor expansion about the wall:

$$\psi_{i,1} = \psi_{i,0} + \left(\frac{\partial\psi}{\partial y}\right)_{wall} h + \left(\frac{\partial^2\psi}{\partial y^2}\right)_{wall} \frac{h^2}{2} + O(h^3). \quad (7)$$

Since

$$u = \frac{\partial\psi}{\partial y} = 0, \quad (8)$$

the first derivative term vanishes. Also,

$$\omega_{wall} = \left(\frac{\partial^2\psi}{\partial y^2}\right)_{wall}. \quad (9)$$

Hence

$$\psi_{i,1} = \psi_{i,0} + \omega_{wall} \frac{h^2}{2}. \quad (10)$$

Rearranging,

$$\omega_{i,0} = \frac{2(\psi_{i,1} - \psi_{i,0})}{h^2} \quad (11)$$

Since

$$\begin{aligned} \psi_{i,0} &= 0, \\ \omega_{i,0} &= \frac{2\psi_{i,1}}{h^2} \end{aligned} \quad (12)$$

### 3 Top Moving Lid

The top wall is located at

$$j = ny - 1$$

and the first interior point is

$$j = ny - 2.$$

The lid moves with velocity

$$u = U_{lid}. \quad (13)$$

Expanding the interior node about the wall:

$$\psi_{i,ny-2} = \psi_{i,ny-1} - \left( \frac{\partial \psi}{\partial y} \right)_{wall} h + \left( \frac{\partial^2 \psi}{\partial y^2} \right)_{wall} \frac{h^2}{2} + O(h^3). \quad (14)$$

Substituting

$$\left( \frac{\partial \psi}{\partial y} \right)_{wall} = U_{lid} \quad (15)$$

and

$$\omega_{wall} = \left( \frac{\partial^2 \psi}{\partial y^2} \right)_{wall}, \quad (16)$$

gives

$$\psi_{i,ny-2} = \psi_{i,ny-1} - U_{lid}h + \omega_{wall} \frac{h^2}{2}. \quad (17)$$

Rearranging,

$$\omega_{i,ny-1} = \frac{2(\psi_{i,ny-2} - \psi_{i,ny-1})}{h^2} + \frac{2U_{lid}}{h} \quad (18)$$

Using

$$\begin{aligned} \psi_{i,ny-1} &= 0, \\ \omega_{i,ny-1} &= \frac{2\psi_{i,ny-2}}{h^2} + \frac{2U_{lid}}{h} \end{aligned} \quad (19)$$

## 4 Left Wall

The left wall is located at

$$i = 0$$

and the first interior point is

$$i = 1.$$

Since

$$v = 0$$

on the wall and

$$v = -\frac{\partial\psi}{\partial x},$$

it follows that

$$\left(\frac{\partial\psi}{\partial x}\right)_{wall} = 0. \quad (20)$$

Expanding:

$$\psi_{1,j} = \psi_{0,j} + \left(\frac{\partial\psi}{\partial x}\right)_{wall} h + \left(\frac{\partial^2\psi}{\partial x^2}\right)_{wall} \frac{h^2}{2}. \quad (21)$$

The first derivative vanishes, giving

$$\psi_{1,j} = \psi_{0,j} + \omega_{wall} \frac{h^2}{2}. \quad (22)$$

Therefore

$$\omega_{0,j} = \frac{2(\psi_{1,j} - \psi_{0,j})}{h^2} \quad (23)$$

and since

$$\psi_{0,j} = 0,$$

$$\omega_{0,j} = \frac{2\psi_{1,j}}{h^2} \quad (24)$$

## 5 Right Wall

The right wall is located at

$$i = nx - 1$$

and the first interior point is

$$i = nx - 2.$$

Expanding about the wall:

$$\psi_{nx-2,j} = \psi_{nx-1,j} - \left(\frac{\partial\psi}{\partial x}\right)_{wall} h + \left(\frac{\partial^2\psi}{\partial x^2}\right)_{wall} \frac{h^2}{2}. \quad (25)$$

Since

$$\frac{\partial\psi}{\partial x} = 0,$$

we obtain

$$\psi_{nx-2,j} = \psi_{nx-1,j} + \omega_{wall} \frac{h^2}{2}. \quad (26)$$

Thus

$$\omega_{nx-1,j} = \frac{2(\psi_{nx-2,j} - \psi_{nx-1,j})}{h^2} \quad (27)$$

and with

$$\begin{aligned} \psi_{nx-1,j} &= 0, \\ \omega_{nx-1,j} &= \frac{2\psi_{nx-2,j}}{h^2} \end{aligned} \tag{28}$$

## 6 Summary of Wall Vorticity Formulae

For a lid-driven cavity with

$$\psi_{wall} = 0,$$

the wall vorticity values are

**Bottom Wall**

$$\omega_{i,0} = \frac{2\psi_{i,1}}{h^2} \tag{29}$$

**Top Wall**

$$\omega_{i,ny-1} = \frac{2\psi_{i,ny-2}}{h^2} + \frac{2U_{lid}}{h} \tag{30}$$

**Left Wall**

$$\omega_{0,j} = \frac{2\psi_{1,j}}{h^2} \tag{31}$$

**Right Wall**

$$\omega_{nx-1,j} = \frac{2\psi_{nx-2,j}}{h^2} \tag{32}$$

## 7 Physical Interpretation

The velocity on the walls enforces the no-slip boundary condition. This creates large velocity gradients near the boundaries, which generate vorticity. The Taylor series expansion provides a relationship between:

- Wall streamfunction,
- Wall velocity,
- Interior streamfunction,
- Wall vorticity.

As a result, the unknown wall vorticity can be computed before updating the interior vorticity field.